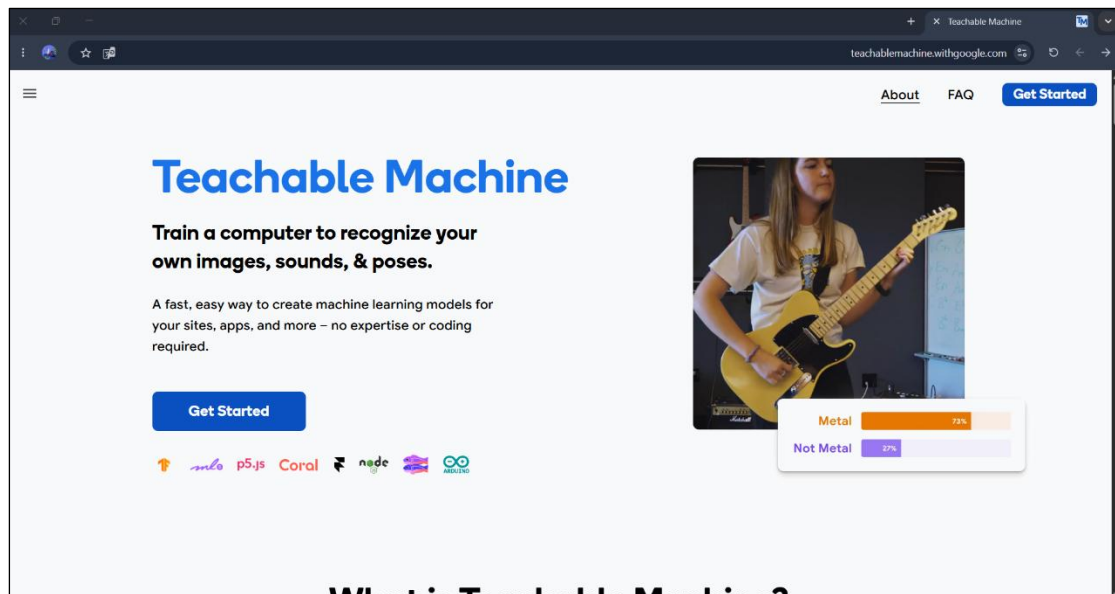


How to Create and Test an ML Image Recognition Model

The following steps explain how to create, train, and test a **Machine Learning (ML) Image Recognition Model** using **Teachable Machine** and **Google Colab**.

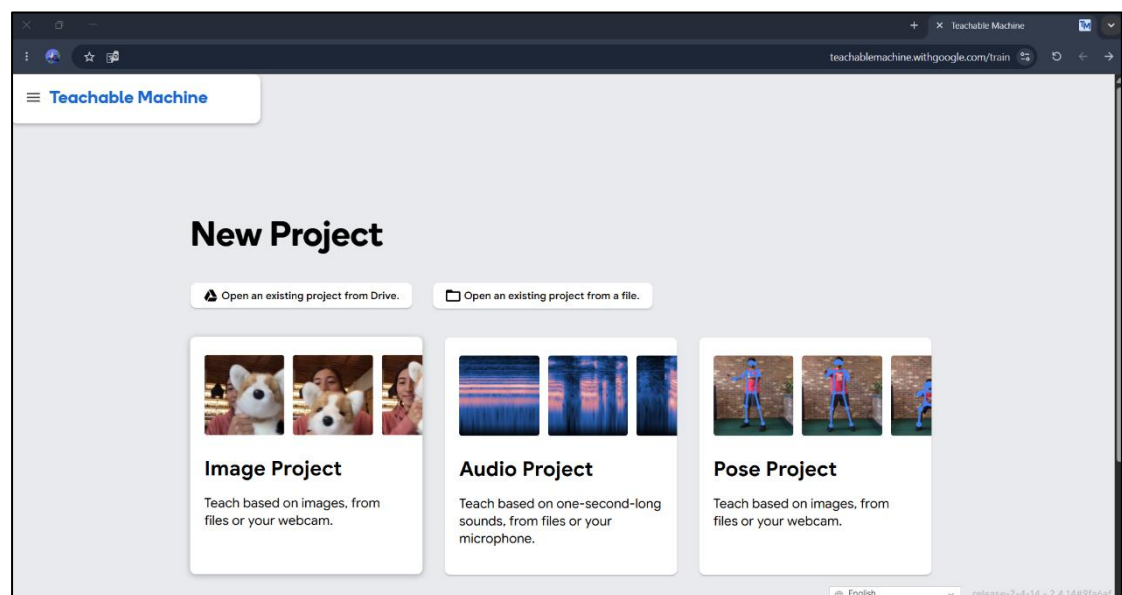
Step 1: Open Teachable Machine

Go to the **Teachable Machine** website.



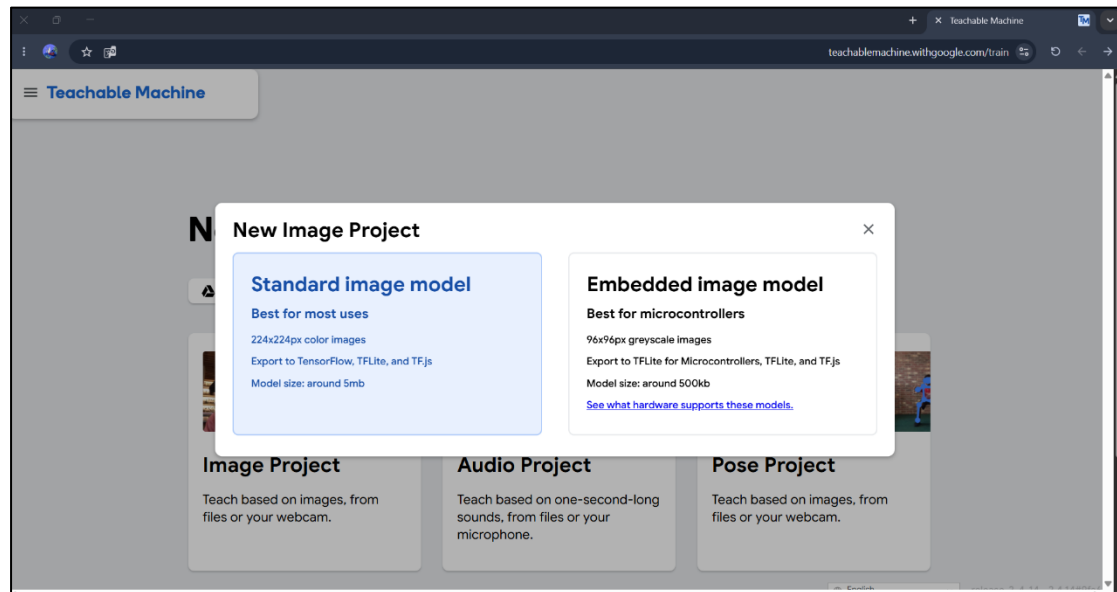
Step 2: Create a New Project

Click on “Create New Project.”



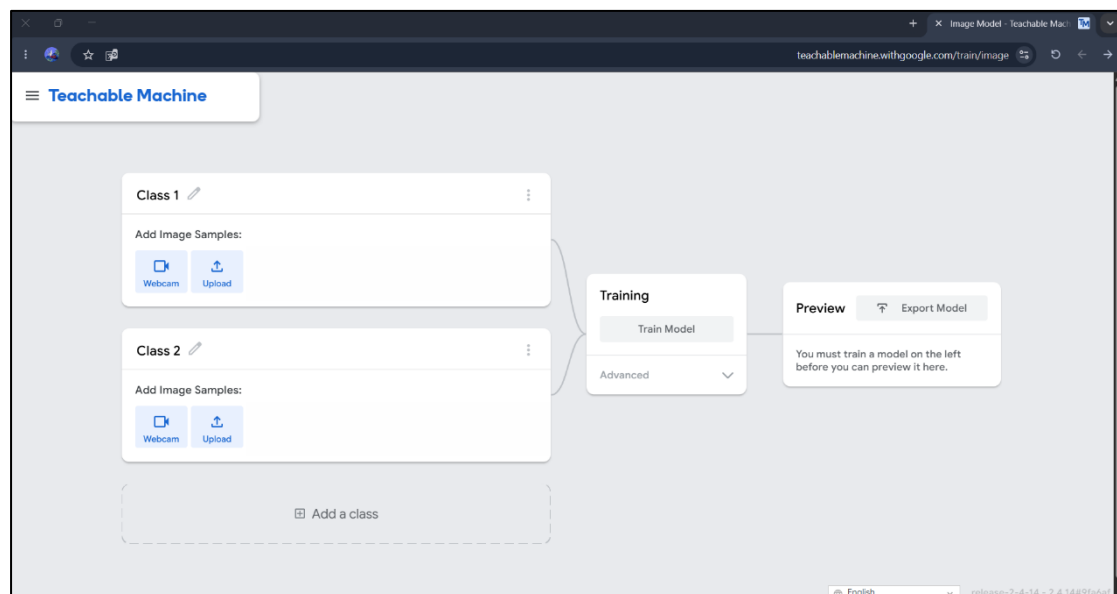
Step 3: Choose an Image Project

From the three available project options, select “Image Project.”



Step 4: Create Classes

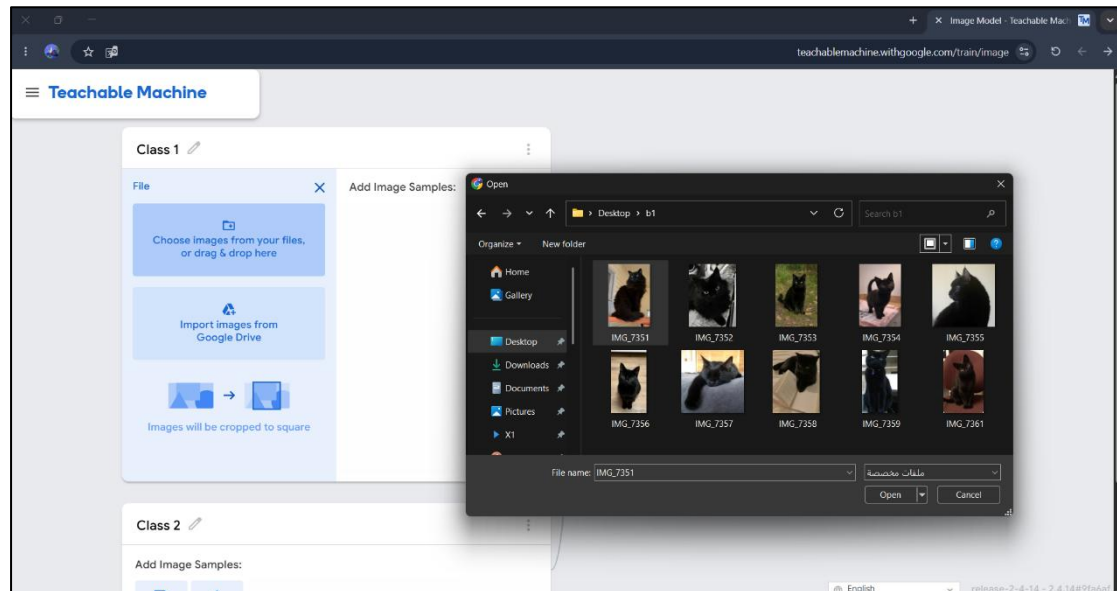
After selecting the image project, a new page will appear. Here, we can create different **classes** that the model will be trained to recognize. For example, we can create a class for **Black Cats** and another class for **White Cats**.



Step 5: Add Images to Each Class

After selecting a class, we can add images that belong to that class.

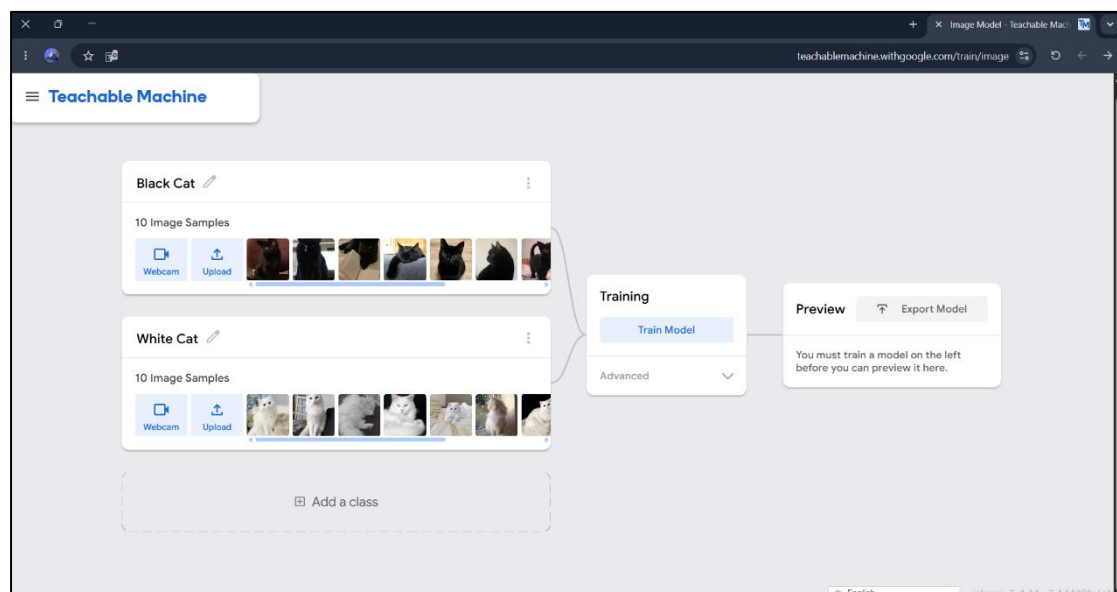
Note: The more high-quality and diverse images we add, the better the model can learn the differences between the classes, which can lead to more accurate results.



Step 6: Train the Model

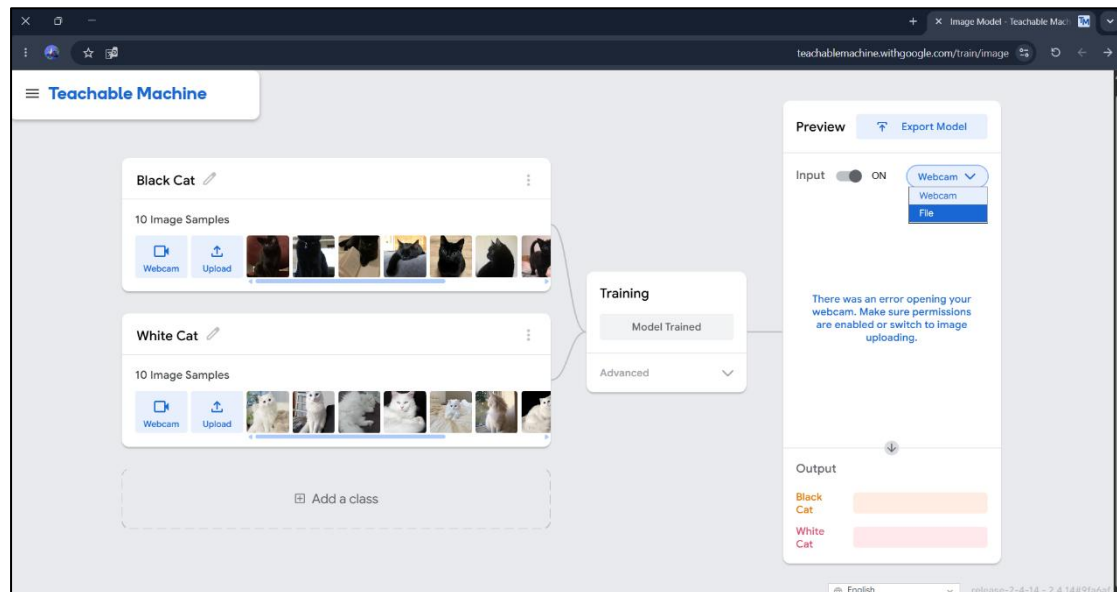
After uploading the images to each class, click on **“Train Model.”**

The model will then analyze the images and learn how to distinguish between the different classes.



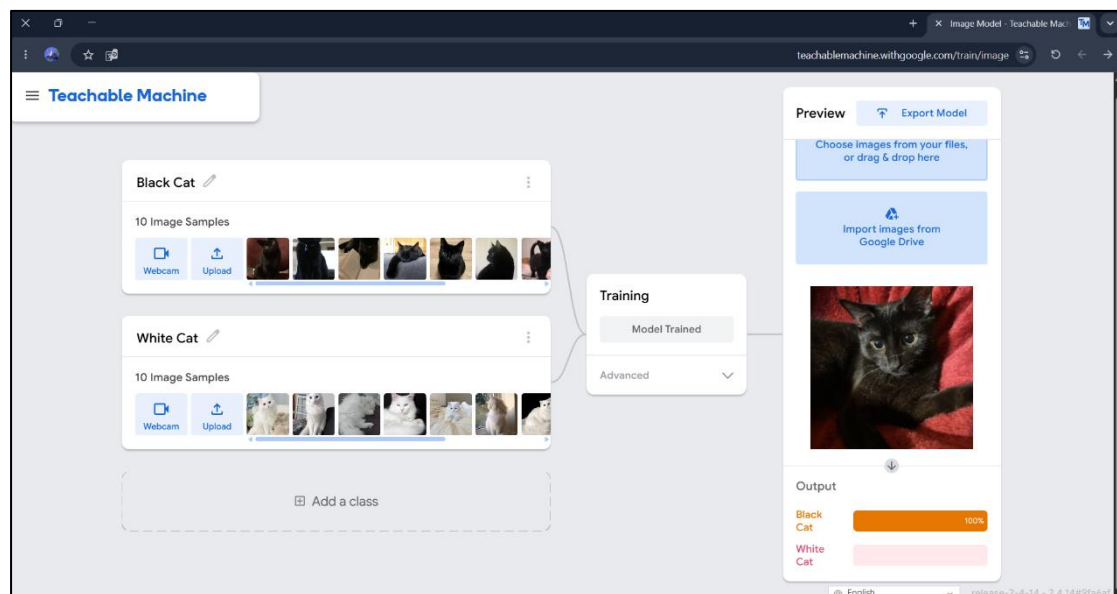
Step 7: Test the Model

Once the model has finished training, we can preview and test the results. We can test the model by providing an input through the **webcam** or by **uploading an image file**.



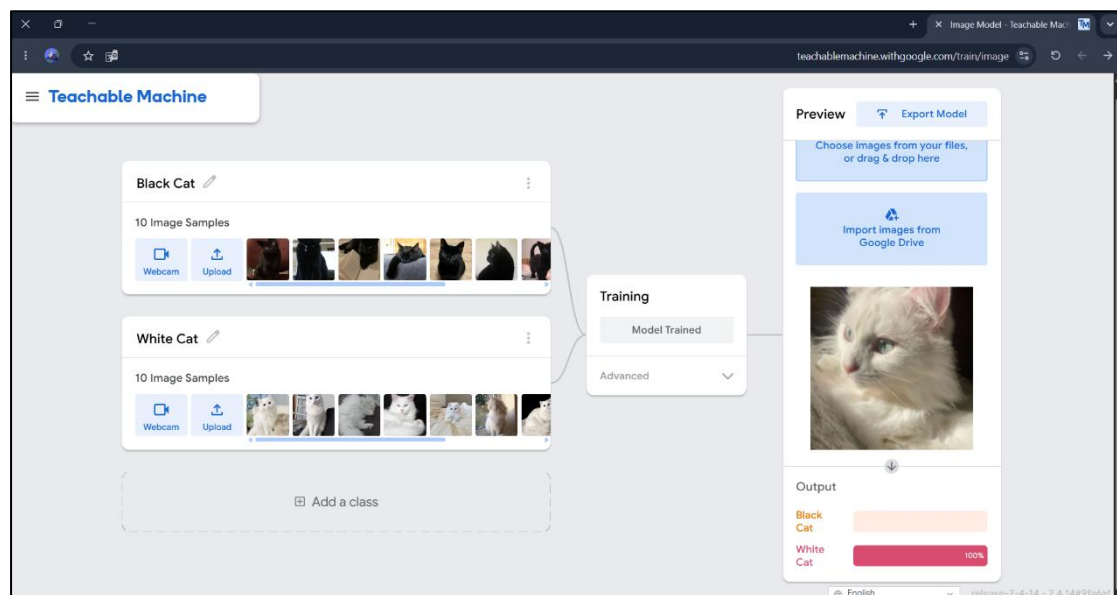
Step 8: Test with a Black Cat Image

After uploading an image of a black cat, the model correctly **classified** the image as a black cat.



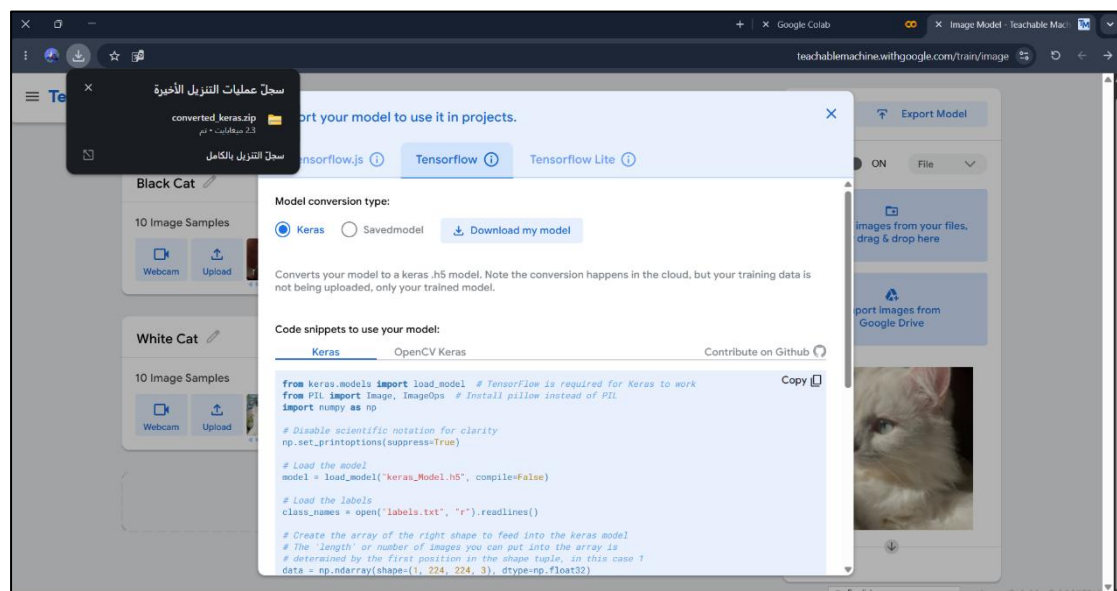
Step 9: Test with a White Cat Image

The same process can be used with an image of a white cat. The model correctly **classified** the image as a white cat.



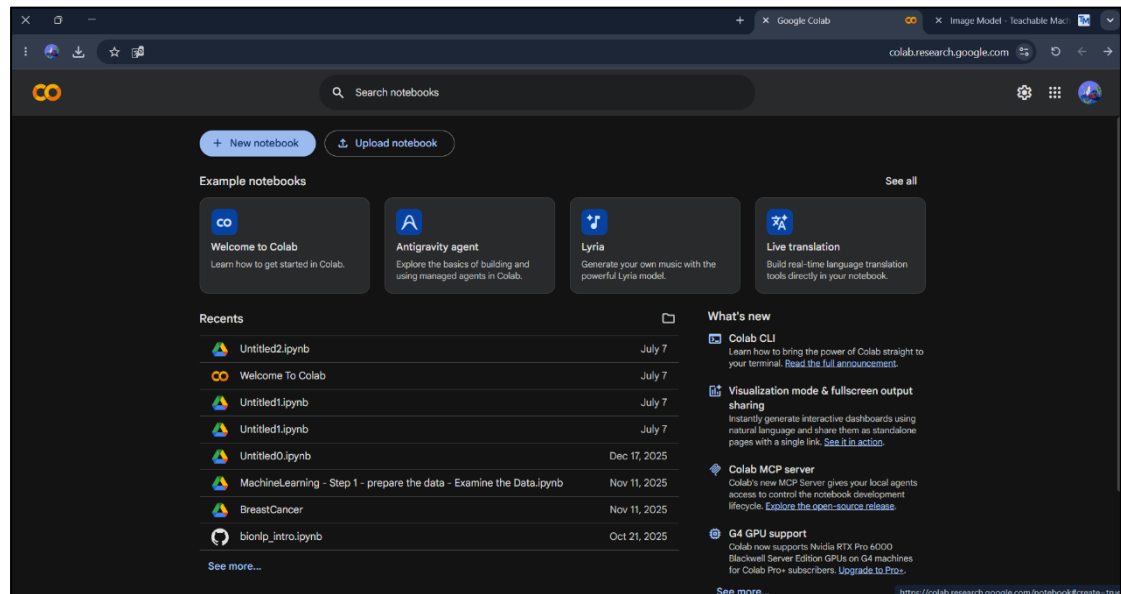
Step 10: Export the Model

After testing the model, click on “**Export Model.**” From the options shown in the pop-up window, select **TensorFlow**. Then, download the trained model and copy the provided code so that we can use and test the model on another platform.



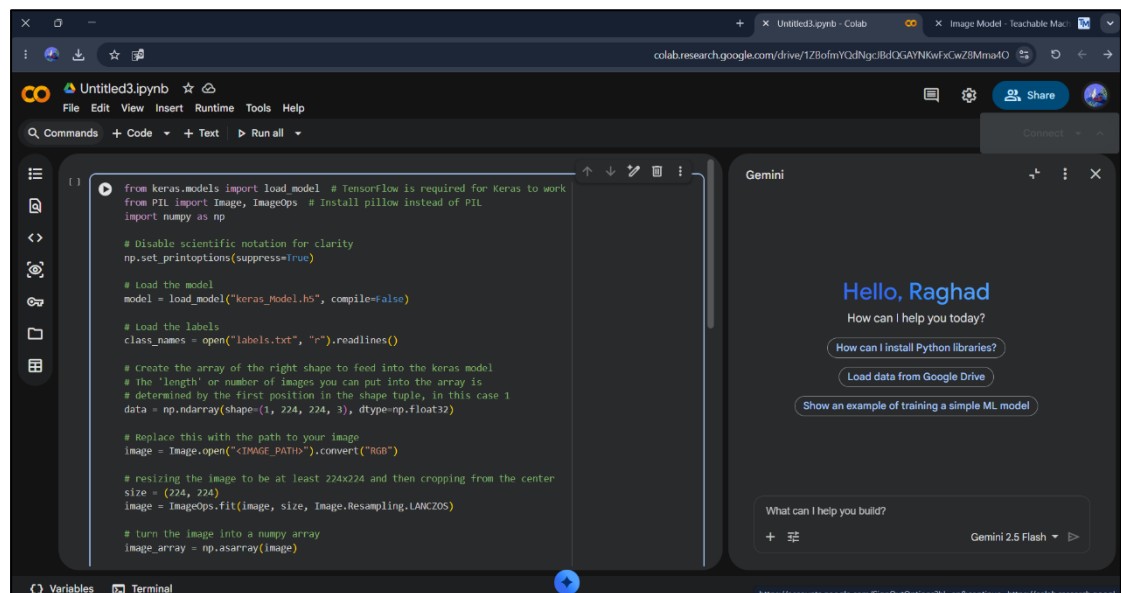
Step 11: Open Google Colab

Next, open **Google Colab** and create a **new notebook**.



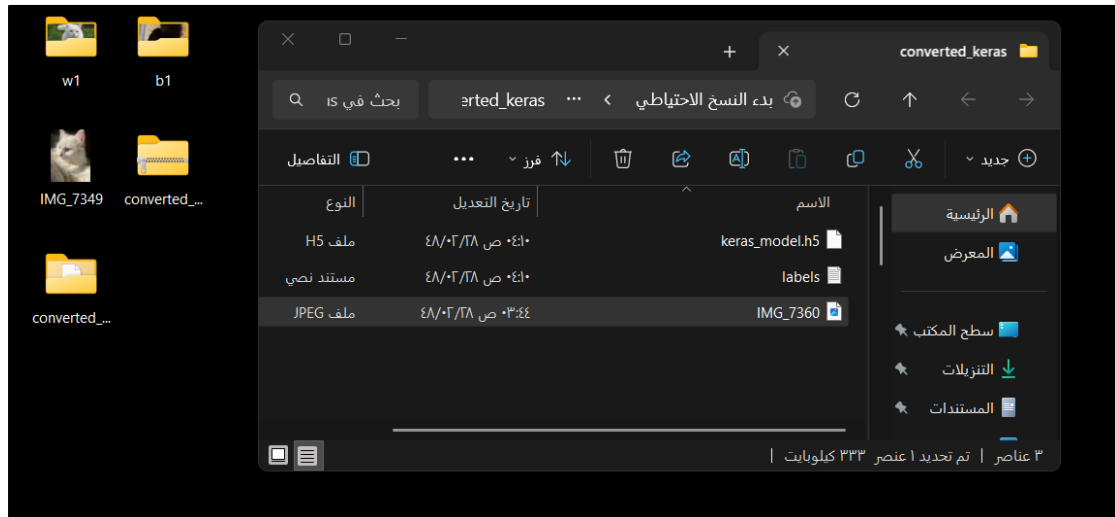
Step 12: Add the Code

Inside the new notebook, paste the code that was copied earlier. Check the code carefully and fix any errors, incorrect paths, or mismatched text if necessary.



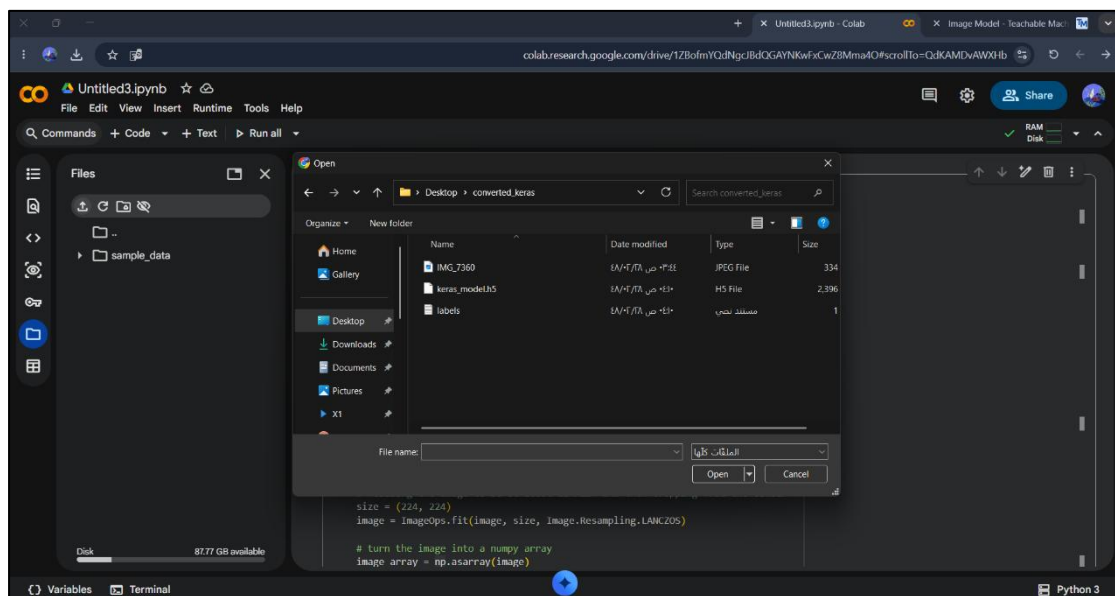
Step 13: Unzip the Downloaded Model

Before adding the downloaded model file to Google Colab, **unzip the file**. The file can also be unzipped directly inside Google Colab by using Python code to extract the contents of the ZIP file.



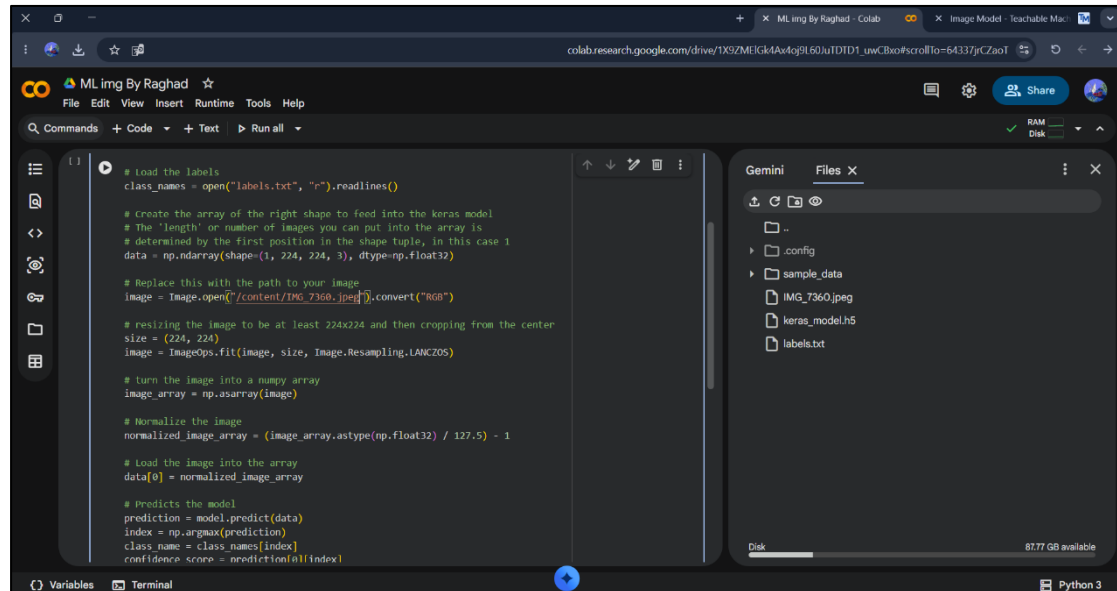
Step 14: Upload the Model to Google Colab

Upload the unzipped model files to **Google Colab** so that they can be accessed by the code.



Step 15: Add the Image Path

It is very important to copy and use the correct **file path** of the image that you want to test. The image path must be correctly added to the code so that Google Colab can find and process the image.



The screenshot shows a Google Colab notebook with the following code in the code cell:

```
# load the labels
class_names = open("labels.txt", "r").readlines()

# Create the array of the right shape to feed into the keras model
# The 'length' or number of images you can put into the array is
# determined by the first position in the shape tuple, in this case 1
data = np.ndarray(shape=(1, 224, 224, 3), dtype=np.float32)

# Replace this with the path to your image
image = Image.open("/content/IMG_7360.jpeg").convert("RGB")

# resizing the image to be at least 224x224 and then cropping from the center
size = (224, 224)
image = ImageOps.fit(image, size, Image.Resampling.LANCZOS)

# turn the image into a numpy array
image_array = np.asarray(image)

# Normalize the image
normalized_image_array = (image_array.astype(np.float32) / 127.5) - 1

# Load the image into the array
data[0] = normalized_image_array

# Predicts the model
prediction = model.predict(data)
index = np.argmax(prediction)
class_name = class_names[index]
confidence_score = prediction[0][index]
```

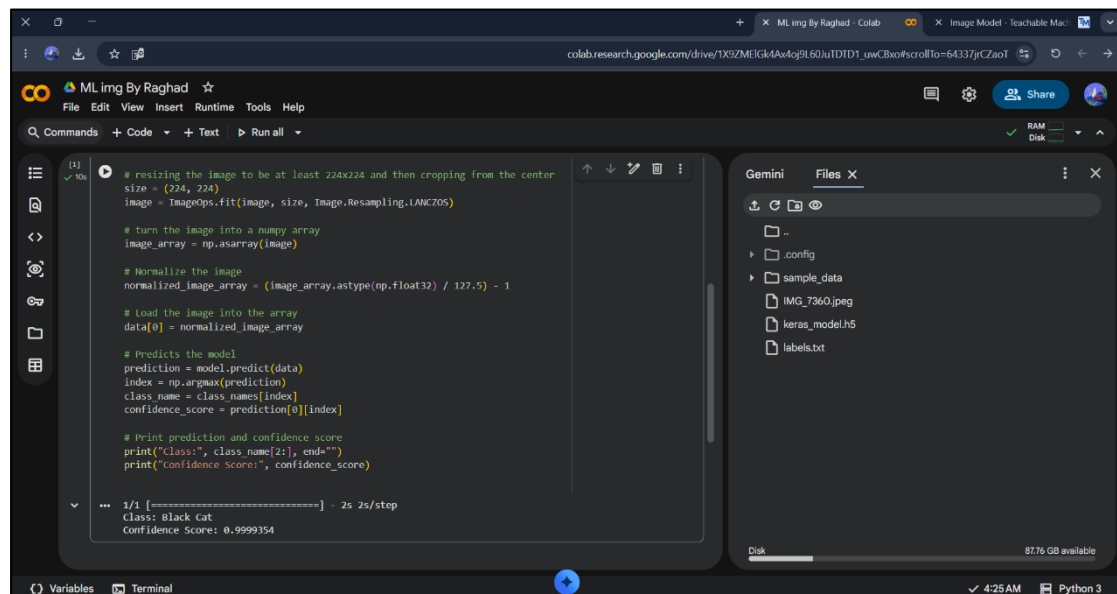
The file browser on the right shows the following files:

- ..
- config
- sample_data
- IMG_7360.jpeg
- keras_model.h5
- labels.txt

The status bar at the bottom indicates "Python 3" and "87.77 GB available".

Step 16: Run the Model

Finally, run the code in Google Colab. If the model correctly identifies the test image, this confirms that the image recognition model is working properly.



The screenshot shows the same Google Colab notebook as in Step 15, but with the code cell executed. The output is displayed below the code:

```
1/1 [=====] - 2s 2s/step
Class: Black Cat
Confidence Score: 0.9999354
```

The file browser on the right shows the same files as in Step 15.

The status bar at the bottom indicates "Python 3" and "87.7s GB available".

By completing these steps, we can create, train, export, and test an **ML Image Recognition Model** and verify whether the model is working effectively and producing accurate results.

The link of Google Colab notebook:

https://colab.research.google.com/drive/1X9ZMEIGk4Ax4oj9L60JuTDTD1_uwCBxo?usp=sharing

H.W by Eng. Raghad Hassan